

Microduck Backflip Showcase: Design and Evidence

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Training Microduck: Backflip Showcase

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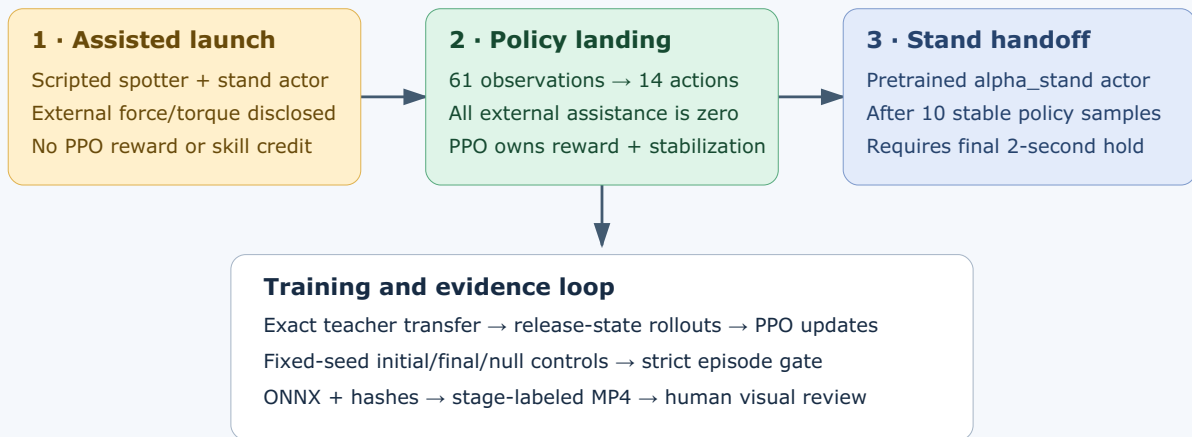
Date: September 8, 2026

Scope: local MuJoCo simulation, RLX PPO, and the five-scenario Studio

This is an assisted demonstration, not an autonomous backflip learned from scratch. A simulated spotter supplies lift and rotation. A pretrained actor initializes the landing policy; PPO updates that actor, but improvement must be established by paired evaluation, not assumed. A separate pretrained stand controller takes over only after the landing policy demonstrates stable support. Neither the launch nor the final stand hold is credited to PPO.

Backflip showcase: who controls what?

Simulation only · assistance, learned control, and pretrained standing are separate



1. What changed

Backflip is the fifth selectable Studio scenario beside Dance, Swing, Running, and Stilts. It uses the existing job queue, saved runs, training telemetry, deterministic evaluation, MP4/contact-sheet renderer, source-bound verification, and ONNX download. The other four environments and rewards are unchanged.

The new Python environment lives in `rlx/rlx/environments/backflip.py`. The physical gate lives in `rlx/rlx/environments/backflip_evaluation.py`. The initializer is in `rlx/rlx/models/backflip_bootstrap.py`. The common CLI is `rlx/examples/ppo_microduck_studio.py`.

The scenario uses the same **61-observation / 14-action** interface. No extra phase, world position, or privileged contact channels are given to the actor. The composite controller knows its stage; the landing actor does not receive that stage as an extra observation.

2. Why the first attempts were rejected

The first API smoke uncovered an actual macOS Objective-C initialization crash when ONNX Runtime was first loaded in forked workers. Backflip now uses the existing dummy vector backend, keeping simulation in the parent process. This avoids unsafe Objective-C fork workarounds. Other scenarios retain their existing backend choices.

The initial torque-only launch released around 4.6 radians with high angular speed. Initial experiments were not accepted merely because training finished:

Experiment	Fixed evaluation	Observed outcome
Behavior-cloned initializer, original gate	Seeds 101–104	3/4 showcase passes
First 100k PPO transitions	Same four seeds and original gate	0/4

Experiment	Fixed evaluation	Observed outcome
BC + 2,000,896 PPO transitions	Strengthened gate, seeds 101–104	0/4
Frozen-normalizer BC + 200,704 PPO transitions	Strengthened gate, seeds 101–104	0/4
Exact teacher transfer, old launch	Strengthened gate, seeds 101–108	4/8
Exact transfer + 200,704 low-rate PPO transitions, old launch	Same eight seeds	4/8
Revised assisted launch, unchanged teacher landing	Strengthened gate, seeds 101–108	8/8
Revised assisted launch, zero-action landing	Same eight seeds	0/8
Revised launch, transfer + 2M PPO	Default seeds 7–22	14/16; rejected
Fresh revised-launch 401,408-step run	Default seeds 7–22	16/16 physical passes
Same fresh run, independent audit	Seeds 401–408	8/8 showcase and 8/8 landing-only passes; PPO refinement fails

The first two rows used an earlier, weaker handoff check and are not directly comparable with the strengthened gate. They are retained to explain the debugging path, not to manufacture a learning curve across different tasks.

The lesson is important: a better-looking reward cannot fix an inconsistent release state or a degraded initializer. We changed the explicitly assisted launch physics and preserved the same landing reward. We also replaced approximate cloning with exact teacher weight/normalizer transfer. The cloning implementation remains available for reproducing the rejected study, but it is not the default initializer.

3. The three-stage controller

Assisted launch

Protocol `spotter-launch-landing-stand-v2` uses a 1.2-second cubic Hermite pitch target from zero to 5.3 radians with a terminal target rate of 5 rad/s. A smooth height target rises from 0.12 m to 0.32 m and lowers toward 0.22 m before release. The spotter acts through MuJoCo generalized external forces, not by teleporting the robot's position or orientation.

The pitch controller uses proportional gain 1.5, derivative gain 0.24, and a 1.3 Nm torque clamp. The vertical controller includes gravity compensation, position gain 90, velocity damping 8.5, and a 15 N force clamp. Lateral velocity damping is 3; roll/yaw angular damping is 0.1. These are **simulator assistance parameters**, not actuator settings suitable for the physical robot.

Release requires rotation at least 5.3 radians and backward angular speed in the range 3.5–6.5 rad/s. Failure to reach that window raises an error; the controller does not silently convert a

bad launch into success. The selected probe produced releases near 5.38 radians, 5.1–5.3 rad/s, and 0.211–0.212 m.

Every external force and torque is cleared at release. The first landing action replaces any queued teacher command in the action-delay buffer. The legacy `assist_torque_norm` field is a generalized-force **zero-detection** channel; `assist_force_n` and `assist_torque_nm` report forces and torques separately in physical units.

Learned landing

During training, `reset` simulates the same real assisted launch, then returns the release observation to PPO. Assisted preparation steps are outside the PPO transition budget and receive no PPO reward. Every collected training action belongs to the landing actor. There is **no stand-policy override in training**.

During the showcase, the landing actor must reach at least 5.8 radians and produce ten consecutive stable samples (0.2 seconds at 50 Hz):

- Both feet contact the floor; no non-foot body part supports the robot.
- Upright projection is at least 0.9.
- Trunk height is at least 0.105 m.
- Gyroscope norm is below square root of 2 rad/s.
- No external assistance is active.

Stand handoff

Only after that pre-handoff evidence does `alpha_stand.onnx` take control. The evaluator independently reconstructs the preceding stability streak and rotation at handoff. The stand controller cannot finish the required rotation on behalf of a failed landing actor. Acceptance also requires the final 100 consecutive samples, or two seconds, to remain stable.

All complete requested episodes must pass. An incomplete trace, missing or nonfinite metric, one failed seed, post-release assistance, or an unsupported handoff fails the skill gate. A pipeline smoke never becomes skill evidence.

4. Reward design: dense, bounded, and observable

The reward is active **only during the landing stage**. Both launch and pretrained stand stages return zero for every reward component.

Let g_z be projected gravity, e_j the offset of joint j from the standard pose, ω the measured angular velocity, v_j joint velocity, and a_j the current action. Define:

```

U = clip((1 - g_z) / 2, 0, 1)
P = mean_j[0.5 exp(-e_j^2 / 0.60^2) + 0.5 exp(-e_j^2 / 0.15^2)]
S = exp(-||omega||^2 / 2)

J = -mean_j[min(v_j^2 / 100, 1)]
D = -mean_j[min((a_j - previous_a_j)^2, 1)]
A = -mean_j[min(a_j^2 / 16, 1)]

r = 3 U + 2 U P + U P S + 0.05 J + 0.02 D + 0.01 A

```

Term / UI weight	Why it exists	Observable input
landing_upright, 3	Dense recovery direction even away from the target	Gravity, indices 3–5
landing_pose, 2	Broad 0.60-rad kernel guides recovery; narrow 0.15-rad kernel distinguishes precise poses	Joint offsets, indices 6–19
landing_settle, 1	Rewards low angular motion only together with upright posture and pose matching	Gyro, indices 0–2; gravity and joints
landing_joint_speed_penalty, 0.05	Discourages unnecessary joint motion without an unbounded impact cost	Joint velocity, indices 20–33
landing_action_rate_penalty, 0.02	Discourages abrupt command changes	Previous action, indices 34–47, and chosen action
landing_action_size_penalty, 0.01	Discourages large commands and actuator clipping	Chosen action

Each positive component is in $[0,1]$ before its weight, and each penalty is in $[-1,0]$. Therefore the total reward is bounded by **-0.08 and 6**. Penalty weights are nonnegative; there is no double-negation path that rewards violations.

The reward deliberately does not use hidden world position, yaw, phase, or a jackpot for crossing a rotation threshold. It also does **not prove touchdown**: an upright airborne pose can score well. Foot support, body contact, height, and consecutive physical stability belong to the independent acceptance gate. This separation is why reward curves alone cannot establish success.

5. PPO and initialization

The stand teacher has the same 512–256–128 actor architecture, ELU activations, 61 inputs, and 14 outputs. The initializer verifies the ONNX graph structure, copies actor weights and its normalizer, and checks native/teacher actions on 9,600 release-rollout observations. A mismatch above $1e-4$ fails initialization. The normalizer is then frozen, so PPO cannot silently change the meaning of the transferred inputs. This is transfer learning, not training from scratch.

PPO optimizes a clipped likelihood-ratio objective. For old/new policies and advantage estimate Adv_t :

```
ratio_t = pi_new(a_t | o_t) / pi_old(a_t | o_t)
surrogate_t = min(ratio_t * Adv_t,
                  clip(ratio_t, 1-epsilon, 1+epsilon) * Adv_t)
```

The optimizer minimizes the negative policy surrogate plus the configured value loss and entropy term. Advantages use discounted rewards and generalized advantage estimation. Clipping discourages excessive changes; it is not a hard guarantee on the size of every update. Approximate KL, clipping fraction, value loss, and actual deterministic outcomes must still be examined.

Reproduction parameters

recipe.json is the machine-readable source for the final run.

Parameter	Value	Purpose and trade-off
PPO environment transitions	401,408	Rounded from 400k to a complete 2,048-transition batch; excludes assisted reset preparation
Environments	16	Diverse reset states in the existing serial vector adapter
Steps per environment / update	128	2.56-second rollout segments for advantage estimates
Minibatches	4	512 transitions per optimizer minibatch
Update epochs	2	Limits repeated optimization on the same rollout
Learning rate	0.000003	Conservative refinement of a transferred actor rather than large early changes
Initial action standard deviation	0.03	Low-noise exploration around a competent controller
Discount factor	0.99	Balances immediate landing control with future stability
GAE lambda	0.95	Existing bias/variance balance for advantages
PPO clipping coefficient	0.2	Limits the clipped surrogate ratio
Maximum gradient norm	0.5	Bounds aggregate optimizer gradient magnitude
Entropy coefficient	0	No extra incentive to inject landing noise
Value coefficient / value clipping	1 / enabled	Critic contribution and clipped value-update objective
Advantage normalization	Enabled	Centers and scales minibatch advantages
Reward normalization	On	Stabilizes critic scale; raw reward is also retained separately
Observation normalization	Frozen transferred statistics	Preserves teacher input semantics
Seed	7	Reproduces training initialization and reset streams
Domain randomization, noise, delay, yaw	Off	First learn and verify a controlled simulation case
Episode horizon	12 seconds	

Parameter	Value	Purpose and trade-off
		Preserved training/showcase horizon; evaluation measures every complete episode
Checkpoint interval	100,000 nominal transitions	Actual saved steps round to complete rollouts

Before PPO, the current initializer collects 16 teacher episodes of 600 landing steps and calibrates discounted-return RMS from those **9,600 measured samples**. It fits the critic for 10 epochs using Adam at $3e-4$ and minibatches of at most 256 observations. The actor is untouched; parity is checked again after critic fitting. Targets include a steady-state bootstrap tail rather than pretending that a time-limit truncation is a physical terminal state. Both actor and critic enter PPO with a fresh PPO optimizer. The calibrated reward scale is then **fixed**, not updated each control step. These teacher calibration and critic fitting steps are not included in the reported PPO transition budget.

Actor observation normalization uses the transferred standard deviation exactly (`epsilon=0`, a very high finite observation clamp). This is distinct from adapting statistics on new rollouts. ONNX export bakes those statistics into the model so raw 61-dimensional observations remain its public interface.

6. End-to-end Studio walkthrough

1. Open the Studio started by `./restart-lab.sh` and select **Backflip showcase**.
2. Open recipe guidance to read which controller owns each stage and what each reward weight means. Start a uniquely named **Default Full** run.
3. Watch the entire training history. Raw completed-episode return and normalized collection reward have different scales and must not be confused. Inspect policy loss, value loss, approximate KL, and clipping fraction.
4. Run deterministic evaluation. Read **every episode**, including pre-handoff stability and rotation, rather than relying on a mean return.
5. Render the rollout. Play the stage-labeled MP4 and inspect the contact sheet. Confirm a real flight/rotation, unassisted policy landing, and stable handoff.
6. Review saved-run evidence. Evaluation/render records are bound to policy bytes and the stand dependency hash. Changing the source invalidates the evidence rather than silently reusing a previous success.
7. Download the landing ONNX if the evidence passes. It is only the middle controller stage. It is not a complete backflip controller or a hardware deployment authorization. The separate Duck Lab teaching panel is not the required second step of the RLX pipeline.

Command-line reproduction

From the workspace root, choose a new run name to preserve previous evidence:

```

cd duck-viewer
node scripts/rlx-dance-api-e2e.mjs --execute \
  --experiment backflip --run backflip-reproduction-001 \
  --profile full --recipe-json ../docs/backflip-showcase/recipe.json \
  --report ../.restart-lab/backflip-reproduction-001.json
cd ..
rlx/.venv-microduck/bin/python rlx/scripts/audit_backflip.py \
  --run rlx/runs/studio/backflip/backflip-reproduction-001 \
  --output rlx/artifacts/backflip-reproduction-001 \
  --seeds 101,102,103,104,105,106,107,108

```

Run the same API driver with `--profile smoke` and a different run name for a four-transition wiring check. A smoke pass does not substitute for Full training, physical evaluation, or visual review.

7. Evidence and learning interpretation

Latest reproduction: calibrated V5

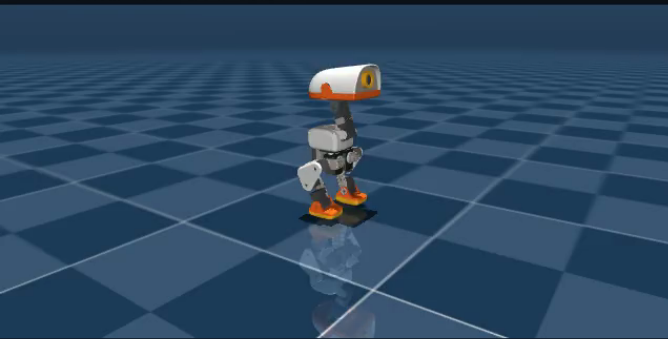
The current recipe reproduces `backflip-e2e-20260908-v5`. Its API training, including initialization, took **128.40 seconds** on this Mac. Evaluation took 4.10 seconds, rendering 8.29 seconds, and export 0.55 seconds. These are measured local job durations, not a performance guarantee for another computer.

V5 passed **16/16** default showcase episodes and **8/8** independent showcase episodes on seeds 801–808. Its exported policy and standalone landing actor passed ONNX parity below $1e-6$. With the stand handoff removed, **7/8** seeds met the final stable-hold check; seed 808 had only 45 stable samples in the final 100. Its paired raw landing return was **2978.57 versus 3409.02** for the initializer, a **12.63% regression**. Consequently, the extended learning audit still reports **FAIL**. The successful three-stage showcase must not be presented as successful PPO improvement or as a reliable standalone landing controller outside its tested handoff contract.

The repaired normalization defect was real: the former cold RMS produced an initial scale near 0.0324 that grew to about 120.42 in one rollout, inflating a raw reward of 3.48 to a normalized reward of 107.44. V5 starts from a measured discounted-return standard deviation of **147.2994** and keeps it fixed. The first collection mean is a finite **0.03612**, without that artificial startup spike. This fixes reward-scale contamination; it does **not** eliminate the separately measured policy drift, and it did not satisfy the improvement gate.

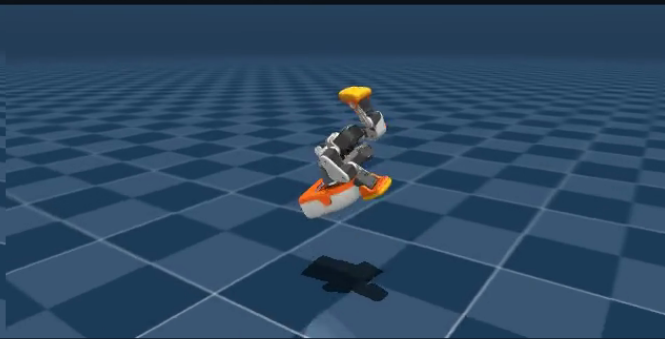
Backflip showcase — actual simulated rollout

ASSISTED SHOWCASE | LAUNCH
Spotter launch > learned landing > pretrained stand handoff



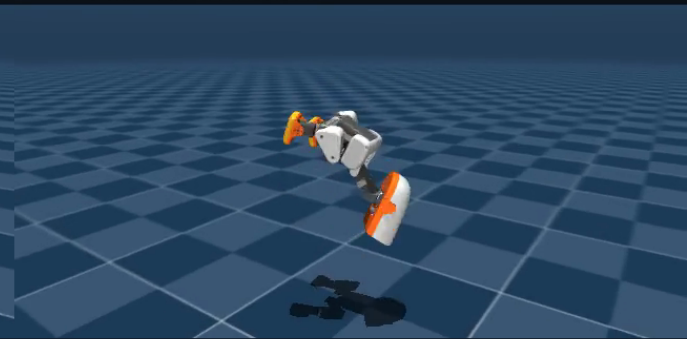
0.00 s · Initial stance

ASSISTED SHOWCASE | LAUNCH
Spotter launch > learned landing > pretrained stand handoff



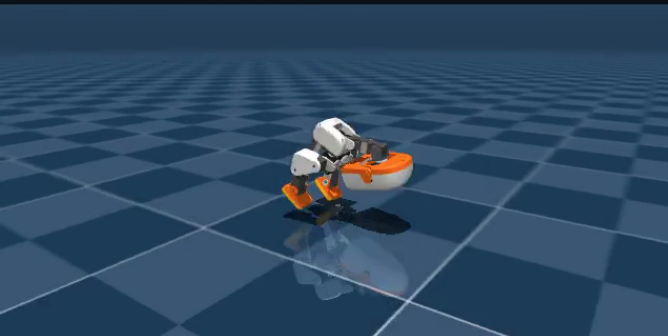
0.68 s · Assisted rotation

ASSISTED SHOWCASE | LAUNCH
Spotter launch > learned landing > pretrained stand handoff



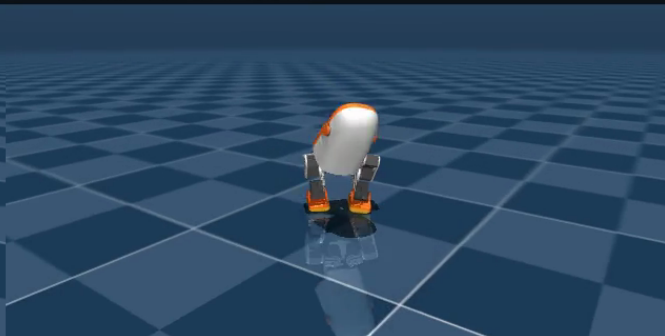
1.00 s · Assisted rotation

ASSISTED SHOWCASE | LANDING
Spotter launch > learned landing > pretrained stand handoff



1.32 s · Policy landing

ASSISTED SHOWCASE | LANDING
Spotter launch > learned landing > pretrained stand handoff



1.68 s · Policy recovery

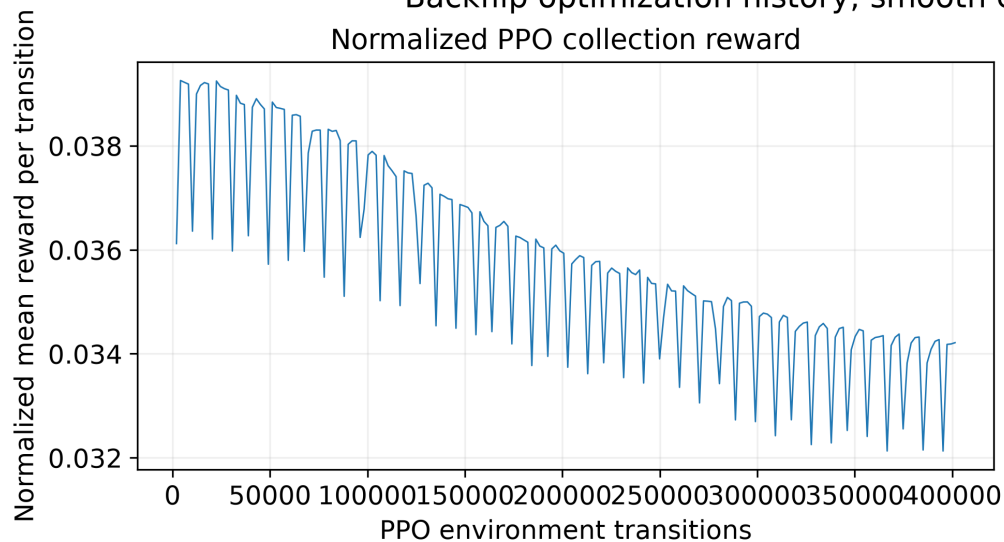
ASSISTED SHOWCASE | STAND
Spotter launch > learned landing > pretrained stand handoff



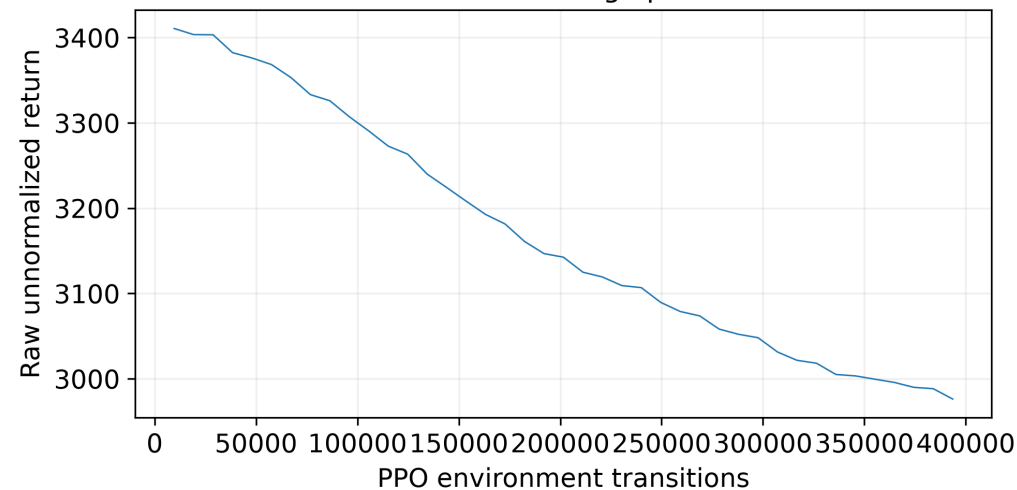
11.96 s · Pretrained stand hold

Backflip optimization history; smooth curves do not establish physical success

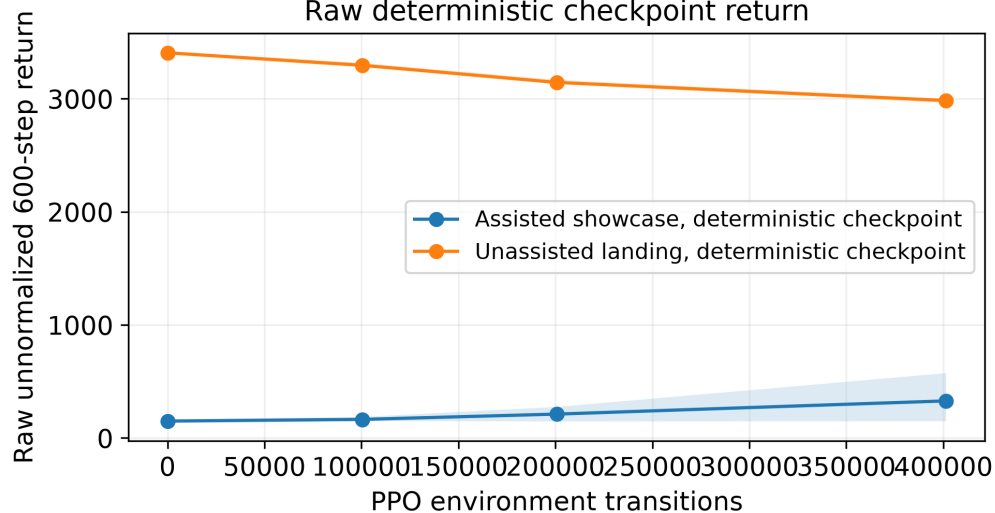
Normalized PPO collection reward



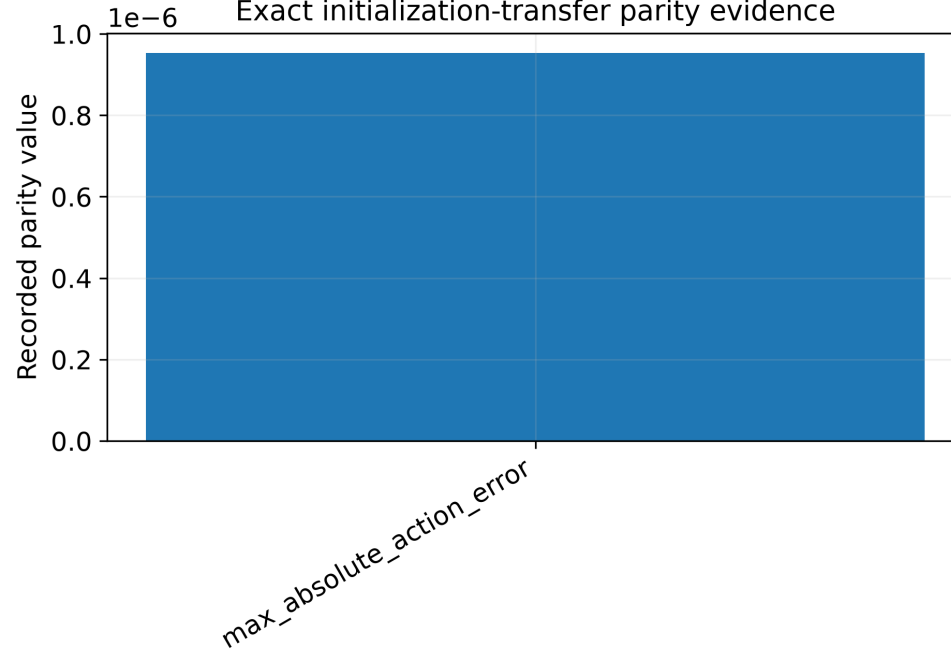
Raw stochastic training episode return



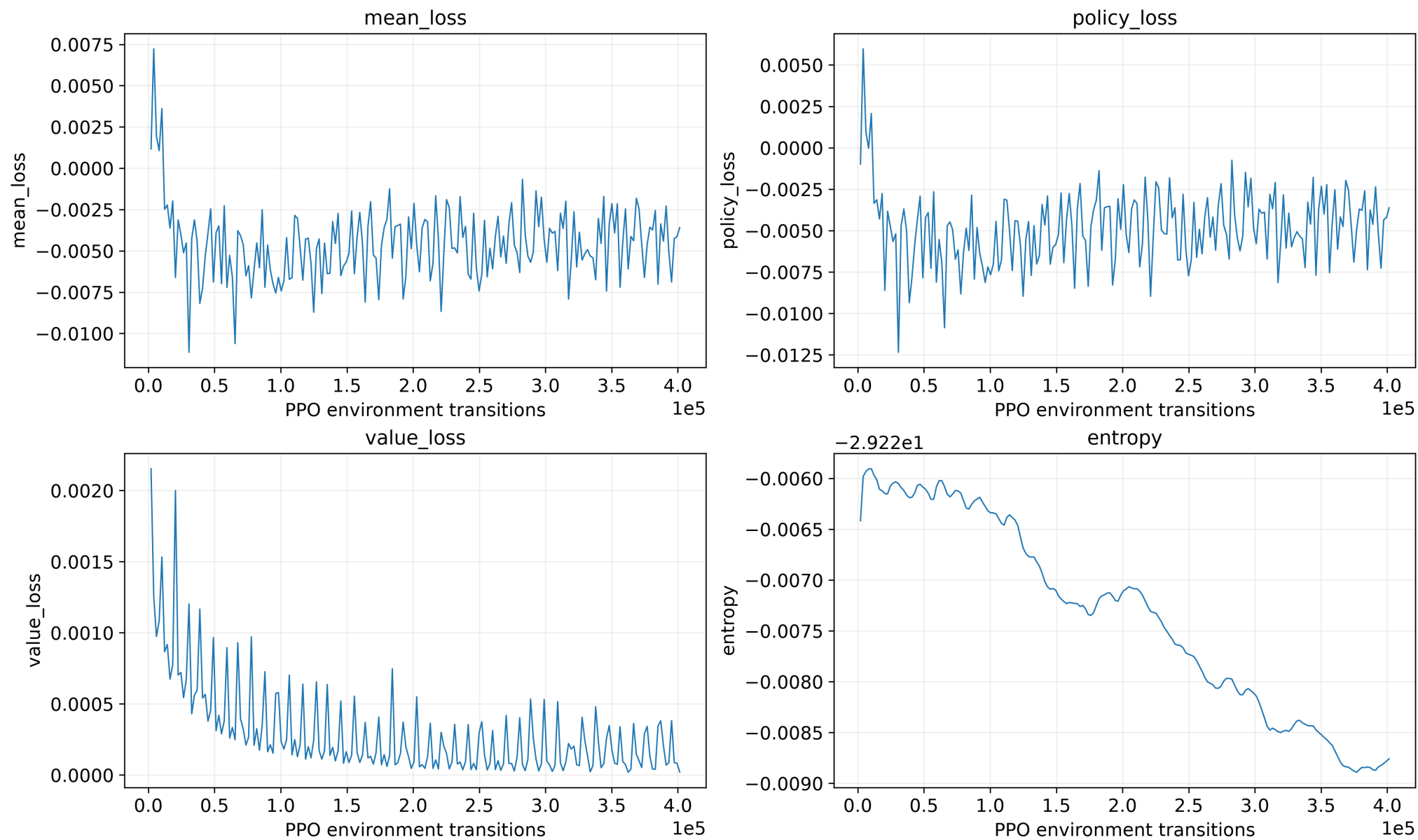
Raw deterministic checkpoint return



Exact initialization-transfer parity evidence

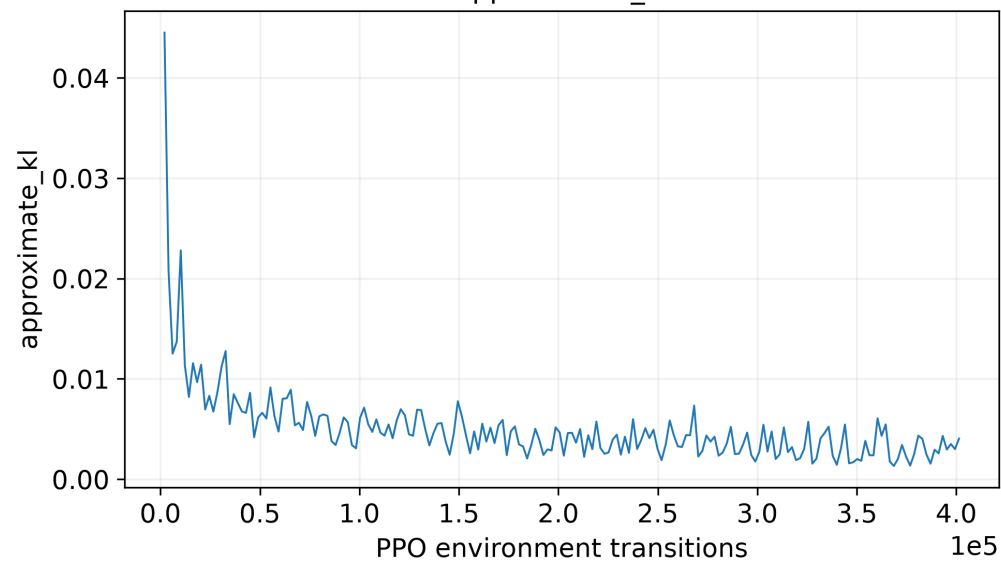


Backflip PPO — full, unsmoothed training history

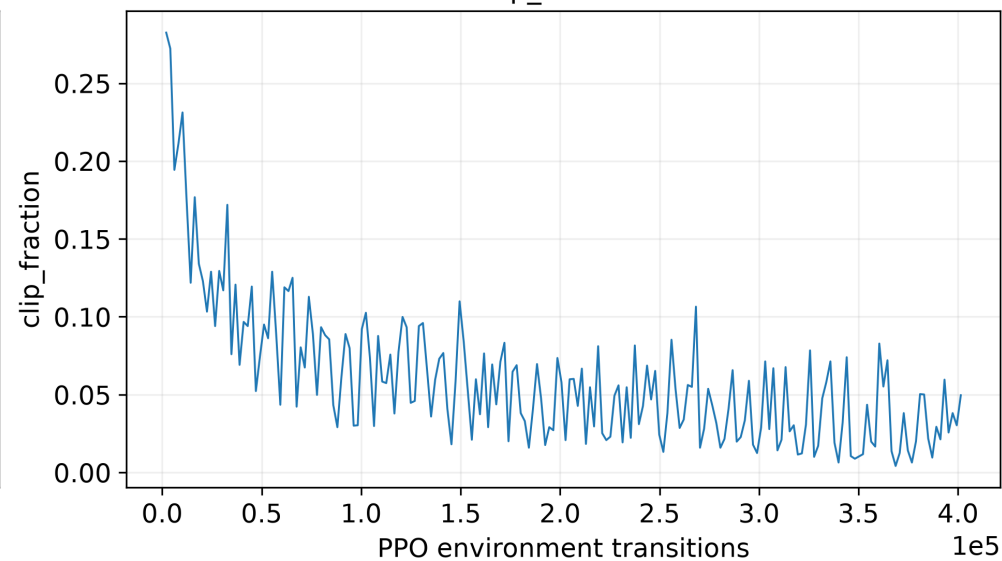


Backflip PPO — full, unsmoothed training history

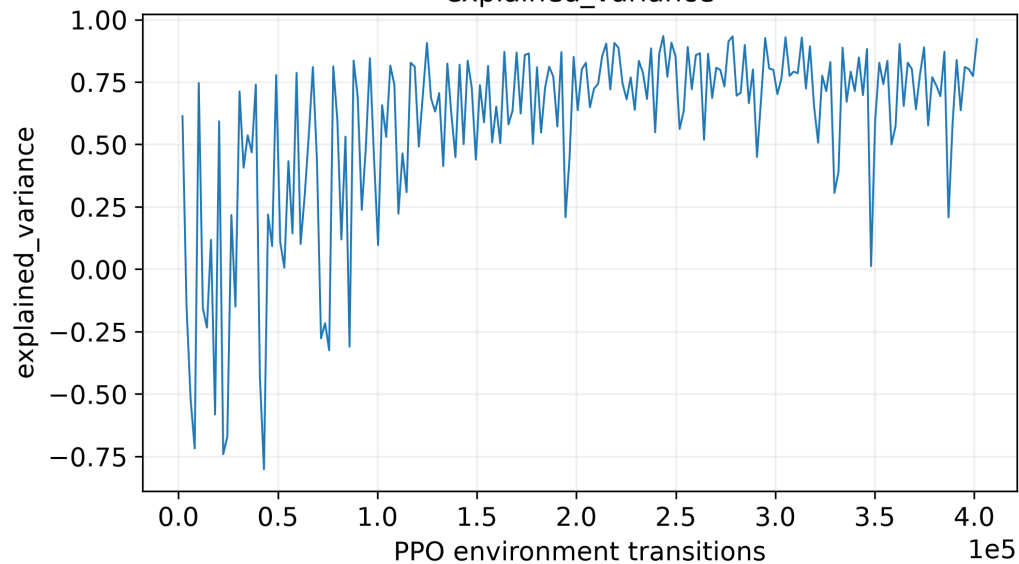
approximate_kl



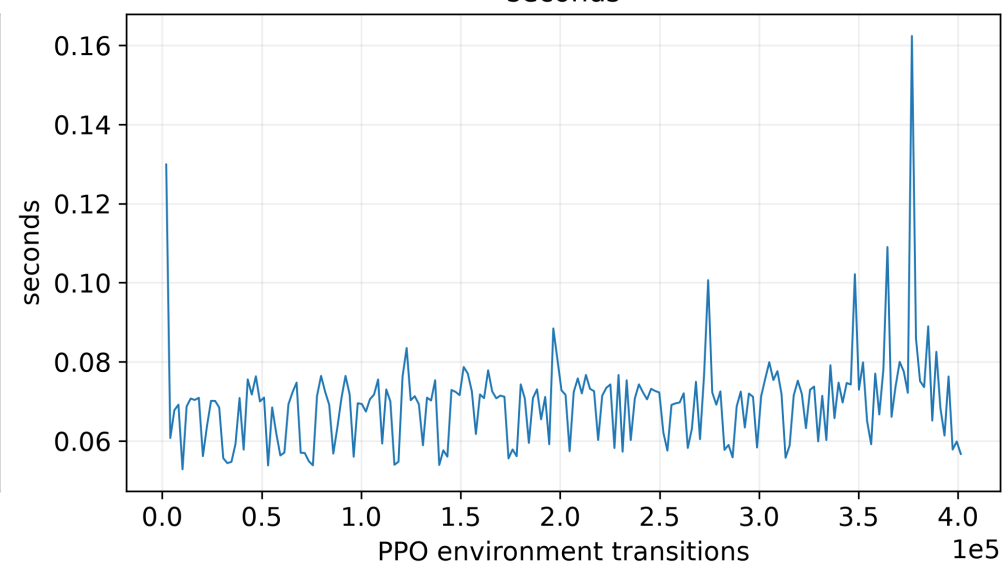
clip_fraction



explained_variance

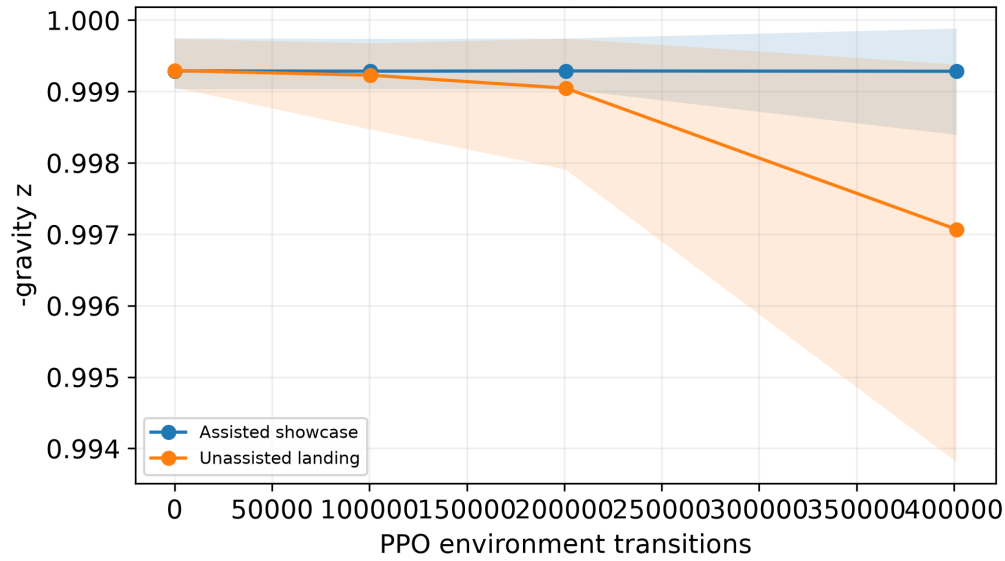


seconds

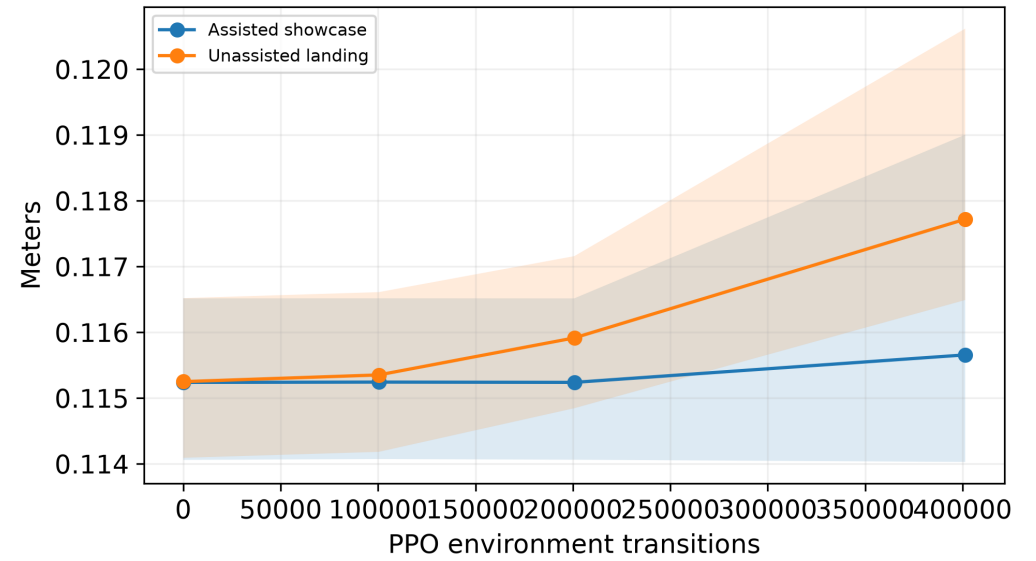


Backflip checkpoint learning curves; acceptance remains seed-wise and physical

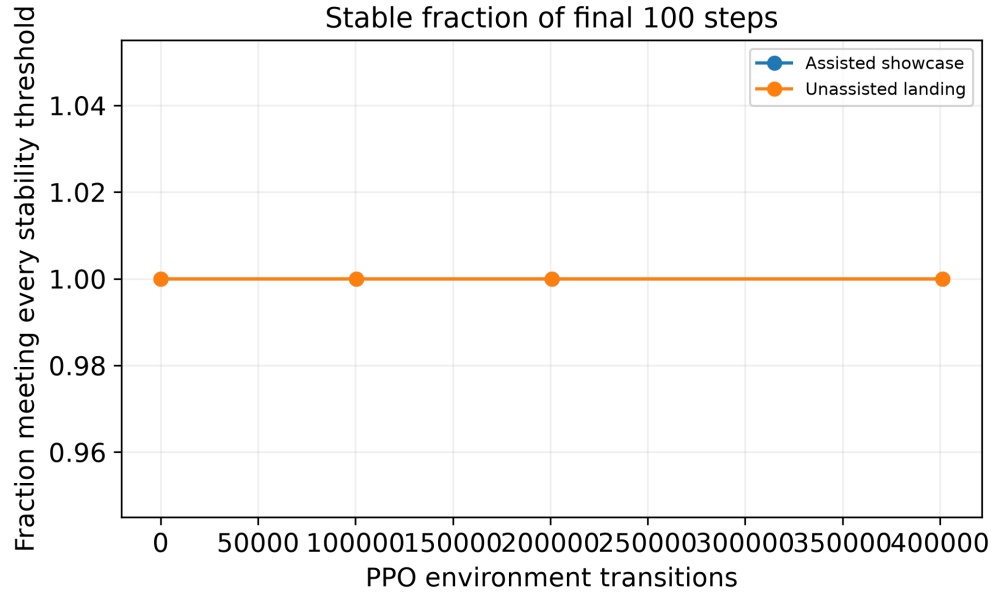
Final upright



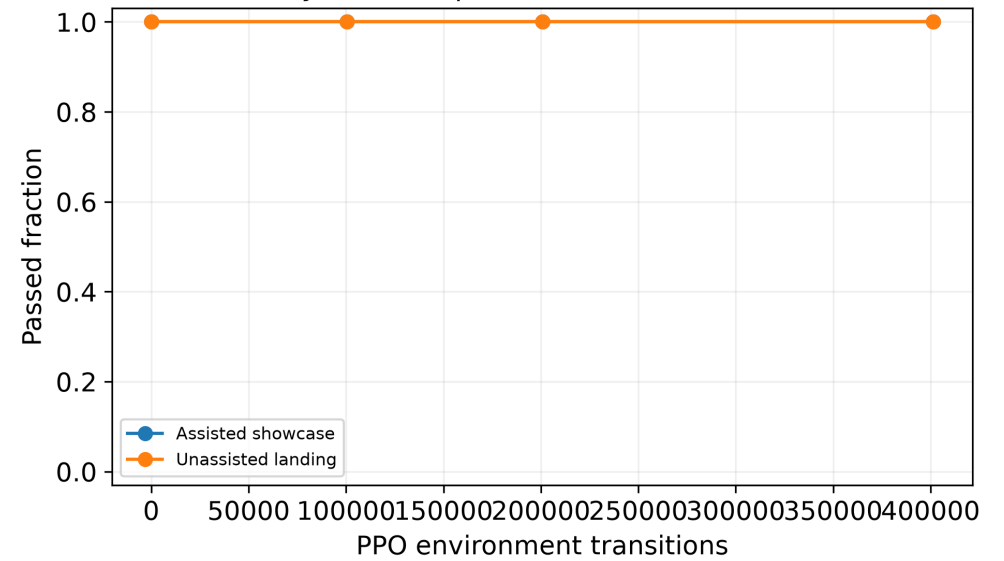
Final trunk height



Stable fraction of final 100 steps



Physical acceptance across audit seeds



Detailed latest evidence lives in `rlx/artifacts/backflip-e2e-20260908-v5/`. Its MP4 is `rlx/runs/studio/backflip/backflip-e2e-20260908-v5/render/ep0.mp4`. The plots are 300 dpi and printed on landscape pages for readable labels.

Predecessor V4: physical success without refinement

The reproduced run is `backflip-e2e-20260908-v4`. The real API completed train, evaluation, rendering, and export. Its default evaluation passed **16/16** episodes. A separate audit on seeds 401–408 passed **8/8** showcases and **8/8** landing-only episodes without any stand-controller handoff. Zero actions failed all eight episodes in both protocols.

However, the PPO improvement requirement is not met. The initializer was already a competent landing actor. The final actor has a lower paired landing-only raw return on every audited seed, despite retaining physical success. The audit deliberately reports overall **FAIL**, with physical-success gates true and the PPO-refinement gate false. Do not turn this into a claim that PPO discovered a new backflip or improved the teacher.

Controller, 600-step landing-only test	Stable final hold	Mean raw return	Final gyro norm
Exact initializer	8/8	3417.42	0.0100
PPO actor after 401,408 transitions	8/8	3025.96	0.0301
Original stand teacher	8/8	3417.42	0.0100
Zero actions	0/8	1942.86	0.0001

The final actor loses **391.46 raw return, or 11.45%**, relative to its initializer. The zero actor's low gyro does not mean success: it lies at about 0.0316 m height with upright projection about 0.135. The physical gate rejects that quiet failure.

An independent fixed-noise comparison on seeds 301–303 also regressed: initializer 3443.88 versus the 401,408-step actor 3030.70 under identical Gaussian action noise with standard deviation 0.03. This is not just a deterministic-versus-stochastic evaluation mismatch. Most loss is in pose and settling; upright reward remains near saturation. Targeted checks found no reproduced PPO sign, GAE, buffer alignment, or MLX action-gradient bug. The supported explanation is cumulative actor drift from a strong transferred policy; critic/advantage error is a plausible contributing factor, not a proven root cause.

The mean raw **showcase** return is not an appropriate learning comparison: successful actors hand off and stop collecting reward, whereas unsuccessful actors can keep collecting small rewards for the rest of the horizon. Use paired **landing-only** returns and independent physical criteria instead.

The sheet prioritizes the first three seconds, so it shows the airborne rotation rather than skipping it between widely spaced frames. It then shows the six-second and final standing states. The full MP4 is retained at `rlx/runs/studio/backflip/backflip-e2e-20260908-v4/render/ep0.mp4`.

These are the actual curves, including the decline. Reward normalization changes scale early in training; the raw episode-return panel separately reveals policy-quality regression. No data were fabricated or smoothed into a success story.

The loss and reward figures are rendered at 300 dpi. The lifecycle diagram is vector SVG. Per-step NPZ arrays, full JSON gate results, normalizer/model parity, source hashes, and raw telemetry remain under `rlx/artifacts/backflip-e2e-20260908-v4/`. The detailed machine verdict is `audit.json`; its `ppo_refinement` section preserves each paired seed delta.

Additional rejected learning ablation

A separate 401,408-transition trial started from random actor weights, without teacher weight or action distillation. It used three-second landing rehearsals, standard deviation 0.15, learning rate $1e-4$, and four update epochs. Mean raw training episode return increased from about 477 to 589, but physical evaluation passed **0/8** seeds (501–508): the actor learned a low crouch, with mean trunk height about 0.054 m. This is evidence of reward optimization, not the requested landing ability. It is not selected as the Studio demonstration.

An optional experimental CLI weight, `landing_teacher_action_penalty`, is off by default. It penalizes deviation from the stand teacher's action on the same observation using $-w * (1 - \exp(-\text{mean}(\text{action_error}^2) / 0.1^2))$. This is bounded in $[-w, 0]$, is applied only to the landing actor, and introduces explicit teacher guidance rather than pretending to be unguided exploration. A 200,704-transition trial at weight 1 preserved all four tested physical showcases but still lost **3.93%** paired raw return on seeds 601–604. It was therefore not promoted to the default profile.

A teacher-guided random-actor trial with early fall termination ran 1,001,472 PPO transitions. It reached greater height than the unguided crouching actor, but **0/8** seeds (701–708) met the full gate: non-foot body support persisted. No failed ablation was relabeled as a successful skill or used for the preview.

A further 401,408-transition continuation reduced exploration to 0.05 and terminated after ten consecutive non-foot-contact samples. It also failed **0/8** showcase episodes (901–908). This was a rejected experiment, not a relaxed gate or a promoted default.

UI and regression verification

The live five-scenario catalog, saved V5 run, recipe guidance, source-bound evaluation, and MP4 were tested at desktop 1600×1100 and mobile 390×844. The browser check produced no page errors and no horizontal overflow. All training-changing POST requests are blocked by that browser test; the separate API driver is the real train/evaluate/render/export test.

The five cards share one desktop row and stack on mobile. The Backflip recipe guidance explicitly records the V5 paired-return regression and failed extended learning audit. The catalog badge and evaluation panel describe the physical showcase gate, **not improvement attributable to PPO**.

Five saved scenarios in the desktop catalog. The badge indicates matched physical showcase evaluation and media, not improvement attributable to PPO.

Top: verdict

Middle: measurements

Bottom: media evidence


Evaluation gate


Run evaluation

Smoke checks the pipeline (4 steps); full evaluates the skill. Swing requires all 1,200 steps (24 s); backflip requires all 600 steps (12 s).

Saved evaluation scope: skill. Current recipe edits do not change saved results.

Skill verified in simulation

16 / 16 episodes passed · backflip-e2e-20260908-v5

Acceptance requires every complete episode, not just a high average reward. The verdict is bound to current policy bytes; preview clips are not training evidence.

spotter-assisted launch → PPO landing → pretrained stand-policy handoff. Fresh Full exactly transfers the pretrained stand actor and observation normalizer into the landing student. Acceptance requires at least 5.8 rad of credited rotation before handoff, then 10 consecutive stable PPO landing steps with no spotter, stand override, or body support, followed by at least 2 seconds of stable standing. The stand policy cannot finish credited rotation.

Evaluated source SHA-256:

1b9712902e9006699a0fa7983116ff016c37337065e551fd75d31af2f3cc34e3

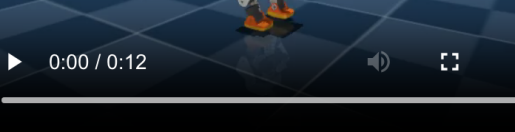
Measurement	Episode minimum	Episode maximum
rotation rad · 16/16 measured	6.2832	6.2832
landing policy steps · 16/16 measured	33.0000	126.0000

landing policy steps · 16/16 measured	33.0000	126.0000
stand hold seconds · 16/16 measured	7.7000	10.1200
min stand height m · 16/16 measured	0.1140	0.1173
stand upright fraction · 16/16 measured	1.0000	1.0000
assist after release steps · 16/16 measured	0.0000	0.0000
Actual-unit channel	Rollout minimum	Rollout maximum
assist force n	0.0000	10.5572
assist torque nm	0.0000	1.1014

► Exact acceptance criteria and per-episode results

► Saved evaluation recipe and raw report

The screenshot shows a web interface for a rollout. At the top, there are two tabs: 'SELECTED RUN' and 'SAVED VIDEO'. Below the tabs is the title 'backflip-e2e-20260908-v5 rollout'. A paragraph of text explains that policy, external dependencies, and media hashes match, and that video settings are a nominal projection of the saved evaluation recipe (randomization disabled). It also notes that this does not visualize every evaluation domain and advises watching the complete rollout after marking it as reviewed. Below the text is a navigation bar with the link 'ASSISTED SHOWCASE | LAUNCH'. Under 'LAUNCH', there are three sub-links: 'Spotter launch', 'learned landing', and 'pretrained stand handoff'. The main content area shows a 3D visualization of a robot on a blue and white checkered floor, with a large, semi-transparent blue sphere in the background.



▶ Open full video

 Open contact sheet

Check	Evidence	Result
● Checkpoint	Safetensors + sidecar	Ready
● ONNX contract	obs[batch,61] → actions[batch,14]	Ready
● Rollout validity	Finite observations, rewards, and actions	Passed
● Rotation	5.857 rad	Accepted
● Visual artifact	MP4 + contact sheet (separate visual review)	Ready
● Video provenance	Source, clip, settings and media hashes	Matched



Render
rollout

▶ Play rollout



Contact sheet

☐

I reviewed the rendered motion

Confirm spotter-assisted launch → PPO landing → pretrained stand-policy handoff, including credited rotation before handoff and 10 support-free landing steps.

Read left to right: the actual evaluation panel, continued in overlapping sections. Inspect episode measurements and controller ownership before accepting the assisted demonstration. This gate does not establish PPO gain.

The full-resolution desktop and mobile screenshots and machine-readable browser receipt are in `docs/backflip-showcase/ui/`. Recipe guidance includes all reward terms, initialization, calibration, protocol thresholds, and the failed learning comparison. Mobile checks use a 390-by-844 viewport.

The print capture uses a 2x pixel ratio and reduced motion. An earlier 2x run timed out while waiting for the mobile evaluation screenshot to stabilize; its failed receipt is preserved in `ui/rejected-high-resolution-capture/`. This was a capture failure, not a failed physical evaluation. All browser assertions remain enabled in the rerun.

Run the reusable browser check from `duck-viewer/`:

```
BACKFLIP_RUN=backflip-e2e-20260908-v5 node scripts/verify-backflip-studio.mjs
```

The read-only readiness check passes all five saved scenarios. Its scope is saved artifacts and service readiness, **not new training of the previous four scenarios**. Frontend tests pass 145/145; lint, typecheck, and production build pass. The Mac-native RLX plus robot-contract suite passes 523 tests with two skips. The separate GPU-stack tests cannot collect because optional `mjlab` is not installed (19 collection errors); they are not reported as passing.

The inspected `PolicyPanel/Lab` files also contain concurrent external edits. Those changes were preserved rather than overwritten by this Backflip task.

Next learning gate

Do not promote a successor on reward curves alone. Require every physical episode to pass, a positive paired landing-only return difference versus the initializer, more improved seeds than regressed seeds, and the same test under fixed action noise. Investigate explicit teacher-preservation regularization, critic warm-up, and observation-derived leg-extension shaping before larger training budgets. These are proposed experiments unless separately measured; they are not claimed solutions.

8. Reproducibility limits and next experiments

- The spotter is a simulator controller supplying substantial external energy. No result establishes an unassisted launch or a safe real-robot backflip.
- Exact initialization starts with an existing ability. PPO benefit must be measured as refinement relative to that baseline, not inferred from an already-successful actor or a smoother graph.
- First vary one factor at a time: release rate, release height, action noise, or body mass. Keep the reward unchanged while testing robustness.
- Use new evaluation seeds after tuning. The launch-study seeds are not a pristine final test set once they influence configuration selection.
- Compare fixed transition budgets and wall-clock time. Count reset preparation separately: it consumes CPU even though PPO does not train on those steps.
- A hardware program requires a separate actuator/latency model, torque and contact safety review, upstream sim-to-real training, and physical safeguards. This local prototype does not supply those guarantees.

References

- Schulman et al., *Proximal Policy Optimization Algorithms* (2017), arXiv:1707.06347, for the clipped policy surrogate.
- MuJoCo documentation, simulation pipeline and `mjData.qfrc_applied`, for explicit generalized external forces.
- Repository training playbook: `microduck_local/AGENTS.md`.
- Repository implementation and regression tests named in this report are the authoritative sources for the implemented controller and evidence gate.